

AI rings turn sign language into speech



These rings translate sign language into speech (Photo Credit: J. Park et al)

Developed at Yonsei University in South Korea, the technology eliminates the bulky gloves and wired sensors used in many existing solutions. AI-powered smart rings that translate sign language into speech in real time could help reduce communication barriers for people with hearing or speech impairments. The wireless system, called WRSLT, uses multiple sensor-equipped rings to track finger movements and hand orientation, with AI interpreting gestures into spoken or written language. Tests showed about 88% recognition accuracy for both American and international sign language, while supporting continuous, sentence-level translation for more natural conversations.

Wireless endoscope sends real-time surgery video using light

Developed at the Fraunhofer Institute for Telecommunications, Heinrich-Hertz-Institut (HHI) in Berlin, Germany, the system integrates battery-powered lighting and communication modules directly into the endoscope. Researchers have developed a wireless endoscope that transmits surgical images via light rather than data cables, paving the way for less cluttered and more hygienic operating rooms. The prototype uses Li-Fi technology, which relies on modulated LED light to deliver secure, low-latency image transmission and supports real-time 4K video during laparoscopic procedures. Tests conducted with surgeons at St. Joseph Hospital Berlin-Tempelhof demonstrated reliable performance, with clinicians favouring the cable-free design over conventional wired systems.

The medical staff of St Joseph Krankenhaus use an OR simulator to test the wireless endoscope prototype in an operating room (Credit: www.fraunhofer.de/en/press)

